Kamino DVI Solver Benchmark

RSL-RL · 300 iterations · three seeds · mean ± 95% Student-t CI

• Isaac-Ant-Direct: tuned DVI is 4.9× faster than current Kamino and 5.0× faster than PR3570 P-ADMM.

• Isaac-Ant-Direct: tuned DVI remains 2.5× slower than MJWarp and remains 1.4× slower than PhysX.

• Isaac-Cartpole-Direct: tuned DVI is 2.0× faster than current Kamino and 2.0× faster than PR3570 P-ADMM.

• Isaac-Cartpole-Direct: tuned DVI remains 1.5× slower than MJWarp and is approximately equal to PhysX.

• Isaac-Velocity-Flat-AnymalD: tuned DVI is 6.8× faster than current Kamino and 7.1× faster than PR3570 P-ADMM.

• Isaac-Velocity-Flat-AnymalD: tuned DVI remains 2.4× slower than MJWarp and remains 1.9× slower than PhysX.

Task	Variant	Envs	Iteration [s]	Total FPS	Reward	Episode length	Success
Isaac-Ant-Direct	Kamino current	4096	2.040 ± 0.028	64263 ± 890	7402.741 ± 1650.401	898.716 ± 1.169	1.000 ± 0.000
Isaac-Ant-Direct	PR3570 tuned DVI	4096	0.413 ± 0.005	317194 ± 3709	10785.473 ± 3745.488	845.261 ± 218.999	0.863 ± 0.539
Isaac-Ant-Direct	PR3570 P-ADMM	4096	2.048 ± 0.023	63990 ± 722	8547.088 ± 4094.777	897.738 ± 3.259	1.000 ± 0.000
Isaac-Ant-Direct	MJWarp	4096	0.167 ± 0.001	785924 ± 4857	8872.347 ± 33.530	876.501 ± 25.822	0.921 ± 0.075
Isaac-Ant-Direct	PhysX	4096	0.304 ± 0.007	433084 ± 9769	6588.875 ± 1341.016	845.561 ± 18.860	0.888 ± 0.032
Isaac-Cartpole-Direct	Kamino current	4096	0.261 ± 0.005	251502 ± 4447	4.942 ± 0.005	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	PR3570 tuned DVI	4096	0.133 ± 0.001	494241 ± 5153	4.950 ± 0.014	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	PR3570 P-ADMM	4096	0.260 ± 0.002	252178 ± 2015	4.946 ± 0.014	300.000 ± 0.000	1.000 ± 0.000
Isaac-Cartpole-Direct	MJWarp	4096	0.087 ± 0.001	749904 ± 9903	4.947 ± 0.009	299.955 ± 0.193	1.000 ± 0.000
Isaac-Cartpole-Direct	PhysX	4096	0.133 ± 0.004	494030 ± 12633	-5.542 ± 0.151	293.953 ± 5.572	0.970 ± 0.009
Isaac-Velocity-Flat-AnymalD	Kamino current	4096	6.057 ± 0.183	16307 ± 423	21.668 ± 0.335	988.969 ± 6.014	0.997 ± 0.002
Isaac-Velocity-Flat-AnymalD	PR3570 tuned DVI	4096	0.889 ± 0.004	110716 ± 450	21.683 ± 0.509	977.112 ± 13.521	0.995 ± 0.005
Isaac-Velocity-Flat-AnymalD	PR3570 P-ADMM	4096	6.317 ± 0.107	15606 ± 254	21.722 ± 0.597	985.220 ± 3.083	0.997 ± 0.007
Isaac-Velocity-Flat-AnymalD	MJWarp	4096	0.371 ± 0.014	265486 ± 10653	15.859 ± 15.941	978.694 ± 31.812	0.750 ± 1.069
Isaac-Velocity-Flat-AnymalD	PhysX	4096	0.479 ± 0.012	205414 ± 5189	19.574 ± 2.938	986.785 ± 3.059	0.995 ± 0.003

Data quality and failures

• Isaac-Ant-Direct: schema v1.1 success differs from TensorBoard in 15/15 runs and 3556/4500 points; report uses TensorBoard success.

• Isaac-Cartpole-Direct: schema v1.1 success differs from TensorBoard in 15/15 runs and 1600/4500 points; report uses TensorBoard success.

• Isaac-Velocity-Flat-AnymalD: schema v1.1 success differs from TensorBoard in 0/15 runs and 0/4500 points; report uses TensorBoard success.

• Schema validation confirms every required reward, episode-length, and success field exists; this is a value mismatch, not missing data.

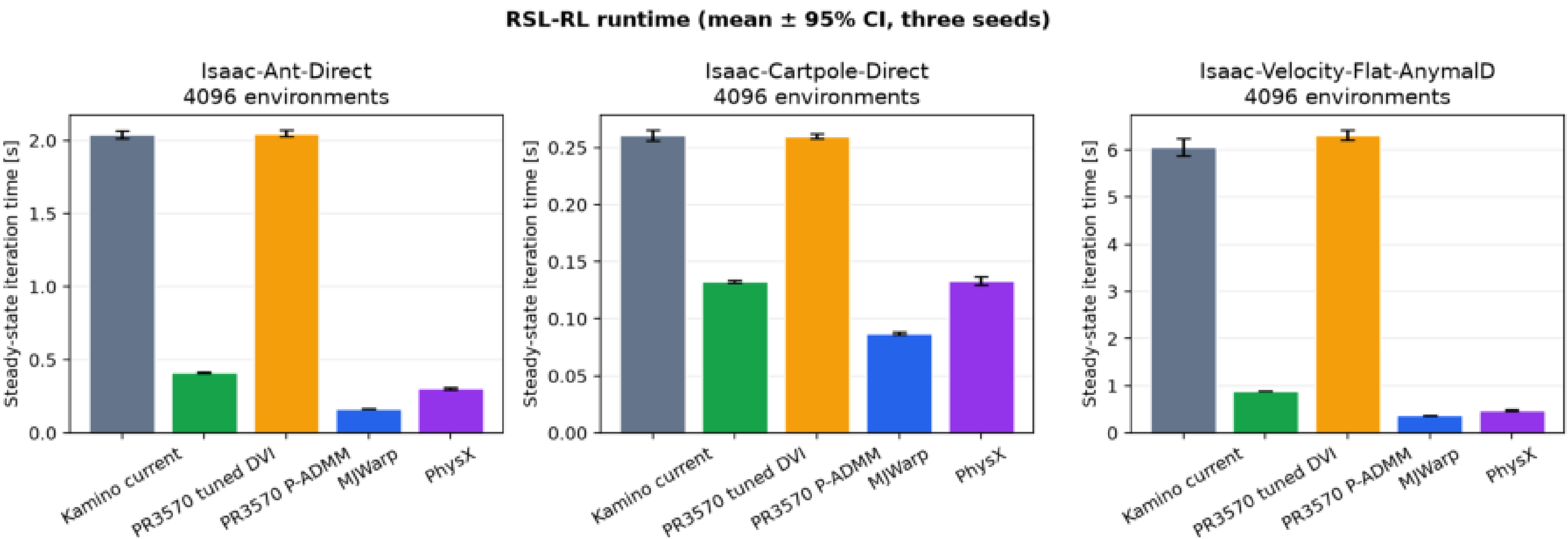
• Isaac-Ant-Direct PR3570 tuned DVI has seed-sensitive weak learning: the three-seed success 95% CI half-width is 0.539; this is not a runtime or stability failure.

• Isaac-Velocity-Flat-AnymalD MJWarp has seed-sensitive weak learning: the three-seed success 95% CI half-width is 1.069; this is not a runtime or stability failure.

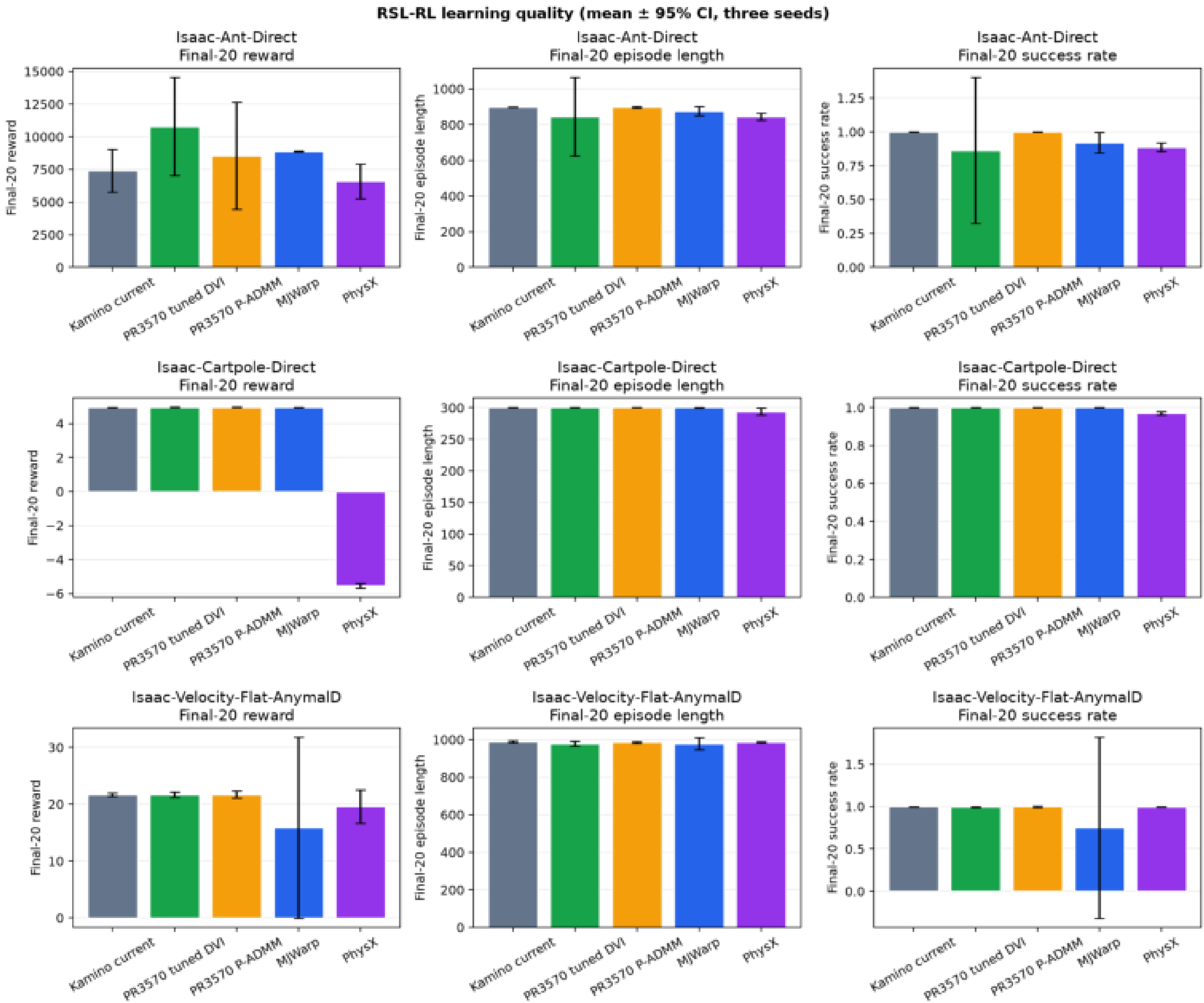
• Legacy campaign source provenance: bundles record 3 distinct commits; 30/45 full runs report versions.git_dirty=true. Runner manifests did not capture exact HEAD, and these runs did not pass the current clean-source check.

• Legacy integrity limitation: TensorBoard event hash was not recorded in 45/45 completed full-run manifests; TensorBoard event files used by those runs were not retained or

Runtime


RSL-RL runtime (mean ± 95% CI, three seeds)
Isaac-Ant-Direct
4096 environments
Isaac-Cartpole-Direct
4096 environments
Isaac-Velocity-Flat-AnymalD
4096 environments
Steady-state iteration time [s]
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX

Learning



RSL-RL learning quality (mean ± 95% CI, three seeds)
Isaac-Ant-Direct Final-20 reward
Isaac-Ant-Direct Final-20 episode length
Isaac-Ant-Direct Final-20 success rate
Isaac-Cartpole-Direct Final-20 reward
Isaac-Cartpole-Direct Final-20 episode length
Isaac-Cartpole-Direct Final-20 success rate
Isaac-Velocity-Flat-AnymalD Final-20 reward
Isaac-Velocity-Flat-AnymalD Final-20 episode length
Isaac-Velocity-Flat-AnymalD Final-20 success rate
Final-20 reward
Final-20 episode length
Final-20 success rate
Kamino current
PR3570 tuned DVI
PR3570 P-ADMM
MJWarp
PhysX